

Purposeful XR: Affordances, Challenges, and Speculations for an Ethical Future

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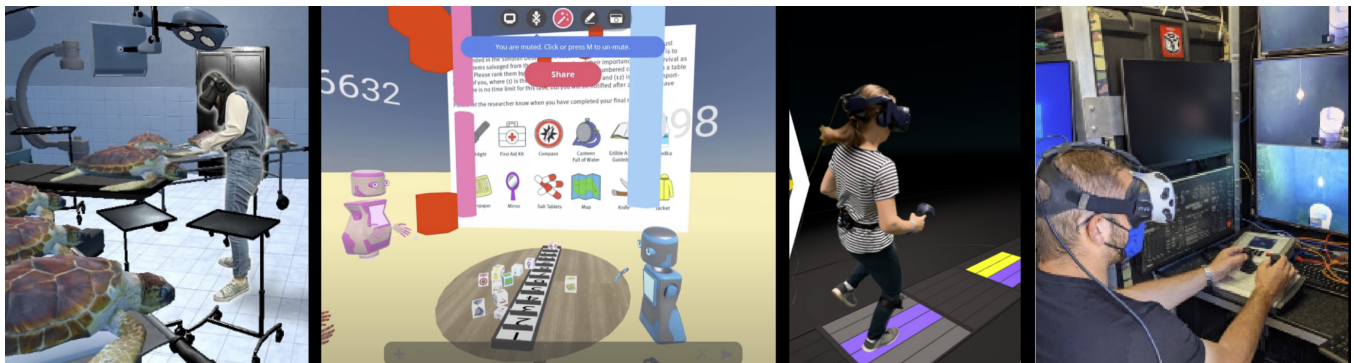


Figure 1: Excerpted figures from diverse XR research for social benefit. *Far left:* A user inspects Loggerhead turtles in a virtual environment [27]. *Center left:* Multiple avatars in a Social VR application use a conversation balance visualization [11]. *Center right:* A user plays a VR exergame focused on leg training [8]. *Far right:* An underwater vehicle pilot uses an HMD for a marine science task [11].

Abstract

Education, healthcare, poverty, and equity are just some of the social problems in which XR researchers leverage augmented reality, virtual reality, and mixed reality to create novel solutions. In this workshop proposal, we intend on gathering XR researchers who are interested in making a positive impact with their research, and to use this opportunity to discuss leveraging unique affordances of XR technology and common challenges. In our motivation, we refer to recent human computer interaction gatherings that discuss applications of XR to social benefit. Then, we present a background of research across XR technologies and diverse social problems to illustrate example affordances and challenges. To attend the workshop, attendees will submit a position paper on a particular XR affordances and how it can help address a social problem, and

optionally submit a demo artifact as a sample for group discussion. The workshop will include a keynote, discussion based activities and demos, and conclude with a speculative exercise, where attendees will work together to describe XR technology in the context of an ethical future.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Mixed / augmented reality**; • **Applied computing**;

Keywords

XR, Mixed Reality, Virtual Reality, Augmented Reality, Affordances, Ethics

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1 Motivation

At the heart of human-computer interaction (HCI), and with it also the CHI conference, is the interplay between technology and society. How can technological innovation help us to address the most pressing societal problems? How can it contribute towards a better future? Extended reality (XR), which has garnered lasting and considerable interest in the community, encompasses all technologies on the reality-virtuality spectrum [23]: virtual reality (VR), augmented reality (AR), augmented virtuality (AV), and mixed reality (MR) technologies [16, 28, 29]. Prior research has used XR applications effectively to address timely social challenges from decreasing vaccine hesitancy via immersive games [32] to providing much needed companionship to older adults facing isolation [2]. However, we believe that the true potential of *Purposeful XR* extends far beyond these innovative uses. To realize this promise, we need thoughtful and intentional research and development, while also mitigating risks for harm that may not be yet fully understood considering the novelty of this medium.

Technological innovations in domains such as health, education, equity and sustainability are being advanced throughout the CHI community. However, as these domains are broad, researchers across various areas of social impact do not have the opportunity to share techniques, challenges, and innovation outside of published work or conferences focused on XR broadly. In 2024, the *Meaningful XR* conference convened to discuss implications of XR applications in such social impact areas. Through collaborative meetups, researchers and designers were able to learn different approaches to solving various challenges, as well as best practices to mitigate harms.

In the CHI community, we have already been coming together to discuss specific XR affordances for social good. At CHI 2024, the *With or Without Permission* Workshop [31] discussed the practical and ethical challenges of site-specific AR for social justice. At a recent Dagstuhl Seminar on *Social XR: The Future of Communication and Collaboration* [3], participants discussed how certain affordances of XR such as enabling immersion and enhancing presence would increase the efficacy and impact social XR applications. Lastly, the CHI 2024 workshop on *Novel Approaches for Understanding and Mitigating Emerging New Harms in Immersive and Embodied Virtual Spaces* [13] discussed some of the potential risk factors of new forms of media, also covering XR as one possible example. At such events, participants discussed specific affordances of XR in the context of achieving an impactful and ethical goal. In this workshop, we will go beyond individual technologies, such as site-specific AR, or application types, such as games. Instead, our workshop will provide the stage for a broad, open, and productive discourse on the various ways XR can be ethically developed as positive avenues towards a better society.

2 Workshop Aims

This workshop has the following four aims:

- *Foster a community of researchers using XR to meaningfully address society's social challenges.* Social problems are bigger than one lab, company, or university. To help alleviate many of the wicked problems in society, individuals of diverse skills and backgrounds will need to work together. This workshop

will focus on researchers investigating the affordances of XR, specifically to leverage those affordances to create a better society. XR has unique affordances outside of novelty, which will help to foster new, beneficial use cases that can expand beyond research conferences.

- *Foster a critical discussion of research and values of XR for promoting social good.* To create purposeful research, researchers need to have a shared knowledge of XR's benefits, and how it can translate across domains. While we recognize that technology alone cannot solve these problems, we can come to a collective understanding of best practices and discuss promising avenues towards a beneficial societal use of these technologies. This workshop will provide a platform for those interested in using these technologies to tackle the current pressing challenges in our society to determine appropriate next steps.
- *Deepen discussion on challenges and potential harms of XR applications.* In addition to the benefits of XR, as a community it is important to have thorough knowledge of potential harms so we do not perpetuate current societal issues. We aim to encourage participants to engage in a critical and open brainstorming session regarding challenges that need to be considered when using XR applications to address societal problems. A proactive consideration and shared awareness of potential unintentional consequences is crucial when applying new technologies for social good.
- *Map out future research direction necessary to further solutions to various social challenges.* Workshops are most beneficial when they contribute to the advancement of future research. Within the domain of *purposeful XR*, We aim to document what future research is necessary to solve social challenges. This can lead to potential collaborations or research ideas to help realize a future without the current social challenges.

3 Background

XR has unique affordances that allow for features that could not exist in traditional, 2D mediums. In this section, we briefly talk about known or already explored opportunities and challenges of XR applications. This overview provides the basis for our workshop in which participants will discuss and brainstorm on further, unknown or underexplored research directions, challenges, harms, and ethical considerations.

3.1 Opportunities

In social VR, multiple users concurrently meet in virtual space with real time voice and chat communications, avatar embodiment and custom interfaces. While this technology currently lacks some important features, such as transmitting facial cues [22], VR can already provide social augmentation that is not possible in 2D videoconferencing [21]. Li et al. [19] created a conversation balance visualization that responds in real time to user's speech, showing interlocutors the quantity of contribution to a conversation and promoting turn-taking. While such 3D social augmentation interfaces show promise, there is still an unaddressed design needs for applications for scientists. Olaosebikan et al. [26] co-designed social VR environments with scientists and determined that there

are common ways that scientists across domains would spatially organize data, explore the data, and leverage VR affordances for cognition. They generated design implications for future social VR technologies that could apply to scientific applications in other XR domains such as DR1: Granular control of the environment, DR2: Rich and diverse communication channels, DR3: Robust and fluid whiteboards, annotation, and note taking, DR4: Efficient and insightful meeting minutes, DR5: Seamless data integration, and DR6: Physical manipulation of artifacts.

Exergames, i.e., “digital game[s] where the outcome [...] is predominantly determined by physical effort” [24], motivate users to be more physically active by combining enjoyable gameplay with strenuous exercises. Such games can be an important step in addressing the growing physical inactivity that has been coined as one of the most pressing societal issues [34]. Already today, XR exergames, like Beat Saber [15] or FitXR [12], are among the best-selling VR applications. However, more research is necessary to ensure that future exergames provide a sustainable player experience that is beneficial for health and wellbeing, and for a wide range of users. Accordingly, a growing body of publications is exploring various aspects of XR exergame-based health interventions. In their CHI 2024 paper, Karaosmanoglu et al. [17] reviewed the current state of XR exergame research. Additionally, they identified various underexplored research directions, serving as a cornerstone for further, targeted research efforts. For example, current exergames mostly focus on upper body movements. Few exceptions address this gap; for example, Cmentowski et al. [8] explored the design of a safe, yet effective and enjoyable XR game to train users’ vertical jump and their lower body muscles. These examples highlight how research can leverage XR affordances to address societal challenges (i.e., lack of physical activity) and how overarching work can identify calls-to-action. In a similar manner, this workshop will likely spark new connections between researchers and lead to interesting new projects that address shortcoming in the current research landscape.

When grounded in education theory, XR has been shown to have positive effects on student learning and motivation. The spatial nature of XR can help students understand spatial subjects such as mathematics [18]. AR can ground digital elements in real locations (location-specific experiences), fostering a deeper connection to a physical landmark. Brewster and Cantave [5] utilized this affordance to shed light on minority stories throughout history. Participants can walk in their local area, and learn how it relates to the untold stories of the hidden heroes. Embodiment, allowing learners to take on different personas, can help them learn by taking a new perspective of understanding. For example, Lindgren and Moshell [20] gave learners the perspective of being an asteroid to learn how gravity and forces interacted. Embodiment has become quite popular for XR, and guidelines have been published to achieve better learning transfer [1]. Lastly, with the ubiquitous rise of smartphones, AR also becomes a cost effective way of bringing these technologies to classrooms.

3.2 Challenges

Within our own research, we often become siloed in our approaches or respective domains. Researchers investigating XR for education

may miss a paper on XR for health, even though the result can be impactful towards education. By gathering researchers across different institutions trying to address social problems, we can share ideas to improve all areas of our research. Additionally, across these domains, XR researchers and practitioners face common challenges, and our workshop activities will encourage participants to identify, discuss, and address these challenges.

One common challenge is that it can be difficult to find contexts in which people use XR regularly in a structured way. By gathering people from different institutions, we can have a discussion on institutional or repeated use of XR technologies to share ideas for distribution and implementation of XR applications.

There are also challenges in making XR technologies widely accessible. Financial barriers and human factors are some obstacles that can prevent these technologies from reaching wider ranges of users including those in the Global South, the elderly and the young, and those with visual, mobility, and cognitive impairments. Through this workshop, we hope to not sideline these users, but embrace designs and ideas that include and empower diverse populations.

Building novel XR applications can involve technical labor and new skill sets. As hardware and software technologies for building XR applications change with each product cycle, XR practitioners have to continuously update skills and evolve technical strategies for development and distribution. One such relevant solution from the research community is Ubiq [14] which attempts to address technical hurdles in creating novel multi-user experiences. At this workshop, we expect participants to trade approaches and tools for development.

Additionally, it is important to note the nuances and limitations to how XR can facilitate empathy. While research by Pimentel and Kalyanaraman [27] show promising results for conservation efforts in getting users to empathize with threatened Loggerhead turtles, Nakamura [25] warns about problematic and spurious forms of empathy that could be generated from VR experiences.

Designing in 3D can require new ways of ideating and creating mockups. For example, with bodystorming [30], groups of designers in VR can use their embodiment to explore how novel interfaces would work. Bressa et al. [4] report on design tools and sketching activities for designing situated experiences. At the workshop, XR practitioners will have the opportunity to discuss strategies for designing.

In addition to human factors lens of viewing methods, XR researchers often use mixed methods in validation. Attendees can convene to discuss the state of the art for such validation methods in XR research. For example, social VR researchers may be interested in analyzing the proxemics of avatars in a virtual environment [33].

4 Organizers

Elizabeth Childs: Elizabeth Childs is a PhD student at Stanford University. She researches how augmented reality can be beneficial towards scientific reasoning and inquiry learning for hands-on science labs. She is creating XR technologies that can be applied in dynamic classrooms while providing teacher the flexible and agency to foster purposeful learning.

Samir Ghosh: Samir Ghosh is a PhD student at University of California, Santa Cruz. He researches how novel multi-user XR interfaces can support scientific sensemaking for applications in wildfire mitigation and marine science. He is a community leader for the Creative Code Collective, an inclusive community to encourage artistic and critical computer programming, where he often teaches hands-on workshops on XR development.

Sebastian Cmentowski: Dr. Sebastian Cmentowski is an Assistant Professor in Game Design and Digital Twins at Eindhoven University of Technology, Netherlands. In the past, Dr. Cmentowski has published extensively on XR-related topics, such as locomotion techniques, non-human character embodiment, and interaction concepts. His recent work focuses on societal applications of XR games, specifically to combat the sedentary crisis through enhanced design and effectiveness of XR-exergame interventions.

Andrea Cuadra: Dr. Cuadra is Assistant Professor of Computer Science at Olin College. She has led the development of inclusive XR technologies for older adults [9, 10], and contributed to research for creating immersive, narrative based learning experiences for younger learners [6, 7]. Her research lies at the intersection of interaction design, inclusivity, and artificial intelligence.

Rabindra Ratan: Dr. Ratan is Associate Professor and AT&T Endowed Chair at Michigan State University. He was lead organizer of the inaugural Meaningful XR 2024 conference, held at Stanford University. His research focuses on social and psychological implications of human-technology interactions.

5 Plans to Publish Workshop Proceedings

The position papers submitted to the workshop will be published together as a collection in ArXiv, the open-access and online repository of research papers. This follows the recommendation to public proceedings in an open access format, and ensure a stable repository for access in the future.

6 In-Person or Hybrid

The mode of the workshop will be fully in-person. A projector, and wifi will be sufficient technology for the workshop. We will need more microphones, but we will bring them to the venue. A power outlet may be required for some XR demos, the organizers can bring necessary converters and extension cables.

7 Accessibility

We strive to make the workshop as accessible as possible. All submissions must adhere to the CHI paper submission guidelines. As the workshop will be fully in person and interactive with no formal overhead projections. Position papers will be both printed and in digital format for participants to access. We will provide writing utensils and paper for brainstorming, but participants can also brainstorm with their laptop if preferred. Participants that are bringing demos will be asked to consider the accessibility of their artifact. Demos that involve head-mounted displays will be asked to provide to ensure that alternate ways of viewing the experience such as videos or photos.

8 Asynchronous Materials

All materials developed during the workshop will be uploaded and accessible to participants to access after the workshop. During the workshop, all materials will be provided via presentation online documents.

9 Workshop Activities

The majority of the workshop will take place in small interactive sessions (See Table 1 for a sample schedule). After a keynote presentation to anchor participants in the XR affordances and how to use them meaningfully, we will dive into discussion using those affordances across various domains. Participants at each table will be assigned an affordance and discuss how the affordance translates across the different domains, ethical considerations of using the affordance on a large scale, as well as the unintentional impacts of furthering the affordance to address the various social challenges. Participants in each group will present their summaries, as well as provide brief introductions of themselves. In session two, we will have a demo session where select participants will demo their work and discuss the various affordances and how they can transfer to other domains.

In session three, the participants will be given the opportunity to change tables and join a new group to discuss technology advances necessary to address the social challenges. Participants will respond to a speculative prompt and discuss what future research is necessary to advance the technology. The focus will be on what needs to happen to further the technology ethically and address challenges in the various social domains. Each table will be given a large sheet of paper and markers to create a diagram or visualization related to the prompt. Again, participants will present their ideas on future technologies, with time for larger discussion.

10 Post-Workshop Plans

A summary report of both workshop discussions will be posted to the website following the workshop. The position papers will be published together in ArXiv. If there is interest among the participants, a communication channel, such as slack, will be established. Otherwise, contact information such as email and school affiliation will be shared with permission of the participants.

11 Call for Participation

We ask prospective attendees to choose a challenge in a social domain and submit a 1-4 page (excluding references) position paper using the ACM Conference Proceedings Preliminary Article Template on how particular XR affordances (e.g., spatial interaction, immersion) can solve that challenge. Good papers will detail a specific social challenge (e.g., increasing science literacy), why solutions inadequately address the problem, and the current barriers with solving the social problem. The rest of the paper will discuss the XR affordance and why it is uniquely positioned to address the social challenge. Papers may reference in-progress or recent work. Lastly, the paper will discuss what technological advances and/or future research is necessary for the affordance to better address the social challenge and which hurdles/harms need to be considered.

Optionally, participants can apply to demo during the workshop. Please submit on the website.

Table 1: Sample Workshop Schedule

Time	Activity	Description
9:00	Welcome	Introduction of organizers and agenda
9:05	Keynote	Talk on XR affordances and their value to tackle social challenges
9:25	Participant Introductions	Participants introduce themselves to the people at their table
9:30	Session 1: Meaningful XR Affordances	Discussion of how affordances are used across different domains, and ethical challenges that arise with the affordance. Participants will present highlights from their discussion to the group.
10:20	Break	
10:30	Session 2A: Affordance Demos	Chosen demos will give a quick introduction to their demo, its affordance and its relation to the social challenge, and any feedback they would like from the group.
10:45	Session 2B: Affordance Demos	Participants will have time to try out demos, ask questions, and give feedback
12:00	Session 3: XR Futures	Discussion of necessary technology and research directions to address social challenges. Participants will present highlights from their discussion to the group.
12:50	Closing Remarks	
13:00	Social	Optional Lunch

While we aim to accept as many attendees as possible, selection of workshop participants will be chosen based on the diversity of uses for their XR affordance, and how well the vision addresses the challenges of the social problem. There are more spaces for participants than demoers, and participants may be selected for the workshop but not to demo. All accepted papers will be published as a collection on ArXiv following the conference.

Please note that at least one author of each accepted submission must attend the workshop and that all participants must register for both the workshop and for at least one day of the conference.

12 Expected size of attendance

We expect to accommodate 15-30 in-person attendees.

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